CTP JOB CARDS PLATFORM

Executive Overview

A real-time digital maintenance management system — transforming fault response, accountability, and asset intelligence across the plant.

Audience: Senior Management **Version:** 1.0 **Year:** 2026

5	2min	4	100%
PRIORITY LEVELS	FIRST ESCALATION	USER ROLES	CLOUD-NATIVE

CONTEXT

The Problem It Solves

In a traditional plant environment, fault reporting and maintenance response suffer from well-known, costly problems:

PROBLEM	BUSINESS IMPACT
No formal fault record — verbal and paper reports are lost or incomplete	No history, no accountability, no basis for analysis
No accountability — unknown when faults were reported, attended to, or resolved	Disputes, delays, inability to measure performance
No real-time visibility — management is the last to know about active faults	Late escalation, production losses compounded
No pattern recognition — no structured data to identify recurring failures	Repeat breakdowns, reactive-only maintenance culture
Slow emergency response — finding a technician requires physical search or radio	Extended downtime on critical faults

The CTP Job Cards platform addresses all of these by digitising the entire process — from fault report to closure — with automated notifications, escalation, and a permanent audit trail.

HOW IT WORKS

The Process — End to End

1. **Fault Reported by Operator**

   An operator discovers a fault and creates a job card on their phone — recording the machine, area, component, fault description, and priority. The job is live in the system the moment it is submitted.

2. **Relevant Technicians Notified Instantly**

   The system identifies which technicians are on site (via real-time geofencing) and their trade (mechanical, electrical, or both). They receive a push notification on their phone within seconds — no dispatcher, no radio call.

3. **Technician Accepts the Job**

   The technician assigns themselves directly from the notification. The operator is notified immediately. Every response — or non-response — is logged with a timestamp.

4. **Automatic Escalation if No Response**

   If no technician responds, the system escalates automatically — first to foremen and on-site managers, then to department managers and the workshop manager. No critical fault can be silently ignored.

5. **Job Tracked to Closure**

   The technician updates the job status and closes it with a completion note. The operator is notified. The full record — fault to fix — is stored permanently and is searchable.

6. **Management Oversight**

   Managers have a live dashboard of all department jobs, technician on-site status, escalation logs, and notification history. They are responsible for enforcing data quality standards.

PRIORITY SYSTEM

Priority & Escalation

PRIORITY	PRODUCTION IMPACT	NOTIFICATION TYPE
P1	No effect — routine or planned work	Standard notification banner
P2	Minor — production continuing	Standard notification banner
P3	Moderate — degraded operation	Standard notification banner
P4	Significant — output reduced	Persistent banner (stays until acted upon)
P5	**Production standing — stopped**	Full-screen alarm (overrides lock screen & DND)

Automatic Escalation Timeline

0 min	2 min	7 min	30 min
Job created — on-site technicians notified	Foremen & on-site managers alerted	Department managers & workshop manager — urgent alert	Further configured escalation

Escalation stops immediately when any technician assigns themselves or acknowledges the job.

Job Card Status

Open → Monitor → Closed

STATUS	MEANING
Open	Job is active — awaiting or being worked by a technician
Monitor	Fault resolved but machine is under observation. Auto-closes after 7 days with no update.
Closed	Job complete and confirmed resolved — full record stored permanently

DATA & RECORDS

What the System Captures

DATA POINT	BUSINESS VALUE
Machine, area, component	Precise asset identification for tracking and history
Fault description	What happened and what was observed
Priority at time of report	Production impact and urgency classification
Who created it, when	Operator accountability and fault timestamp
Who responded, when	Technician accountability and response time (MTTR)
What was done, parts used	Repair record, parts consumption, cost basis
Root cause	Feeds predictive analysis and repeat-failure identification
Full notification log	Complete audit trail — who knew what and when

INFRASTRUCTURE

Security, Access & Infrastructure

User Access Control

Authentication is via company email and password (Firebase Auth), linked to each employee's clock number. Four access levels enforce data boundaries:

ROLE	ACCESS SCOPE
Operator / Technician	Own jobs and relevant department notifications
Manager	Full department visibility, escalation logs, notification history
Admin	Full system — all data, employee management, geofence configuration

Technology Stack

COMPONENT	PROVIDER	KEY PROPERTIES
Database	Google Firebase Firestore	Real-time, offline-capable, serverless, auto-scaling
Push Notifications	Firebase Cloud Messaging	Enterprise-grade delivery, high-volume capable
Cloud Functions	Google Cloud Functions	Serverless escalation logic — no server to maintain
Authentication	Firebase Auth	Email + password, linked to clock numbers
Mobile App	Flutter (Android primary)	iOS and web capable; works offline

✅ **No on-premises infrastructure required.** No server to maintain, no single point of hardware failure. The system scales automatically and is backed by Google's 99.99% availability SLA. All data is encrypted at rest and in transit.

CAPABILITY STATUS

What Is Live Today

CAPABILITY	STATUS
Real-time job card creation and tracking	Live
Push notifications scaled by priority (P1–P5)	Live
Full-screen critical alarms for P5 faults	Live
Automatic escalation (2-min, 7-min, 30-min)	Live
Real-time geofencing and on-site status tracking	Live
Permanent audit trail for every action	Live
Manager dashboard with live department visibility	Live
Offline support with automatic cloud sync	Live
Role-based access control (4 levels)	Live
Notification action buttons (Assign / Busy / Dismiss)	Live
Copper inventory and transaction tracking	Live
Predictive maintenance analytics	Planned
AI fault-finding and manual assistant	Planned

ROAD AHEAD

AI & Predictive Maintenance

The current platform is a foundation. Its long-term value lies in what the structured data it collects makes possible.

PHASE 1

Predictive Maintenance Analytics

Machine learning applied to job card history identifies failure patterns, predicts which assets are likely to fail, and triggers proactive maintenance before breakdowns occur. Shifts the operation from reactive to predictive.

PHASE 2

AI Assistant — Manuals & Fault Finding

A conversational assistant integrated into the app gives technicians instant access to operator manuals, historical repair records, wiring diagrams, and parts specifications — in plain language, on their phone.

Example AI Assistant Queries

"What are the most common causes of overload faults on Conveyor 3?"

"What was done the last three times Machine 7 had a similar fault?"

"What is the correct bearing specification for the FX400 gearbox drive shaft?"

Business Case

BENEFIT	MECHANISM
Reduced unplanned downtime	Predictive alerts trigger maintenance before failure — not after

BENEFIT	MECHANISM
Faster fault resolution	AI-assisted diagnosis reduces time to identify and fix root cause
Knowledge retention	Institutional expertise captured in the system — not lost when experienced staff leave
Lower parts inventory cost	Predictive maintenance enables targeted stock — not blanket over-stocking
Reduced repeat failures	Root cause identified, addressed, and recorded — not just symptoms patched
Technician upskilling	Access to manuals and history on-device reduces dependency on senior staff for routine questions

The quality of AI insight depends entirely on the quality of data entered today. Every job card completed with accurate machine identification, clear fault descriptions, and detailed closure notes becomes a data point that improves predictive accuracy. This is why enforcing data standards at the management level is not an administrative task — it is a direct investment in the long-term intelligence of the maintenance operation.

SUMMARY

What the Business Gets

Speed

Critical faults reach the right technician in seconds. Automatic escalation ensures nothing is silently ignored.

Accountability

Every fault, response, and action is timestamped and logged. Nothing is anonymous or deniable.

Visibility

Management sees everything in real time. No chasing, no guessing, no surprises.

Compliance

A complete, permanent audit trail for every maintenance event — accessible at any time.

Intelligence

A growing data asset powering predictive maintenance and AI-assisted operations as the system matures.

Zero Infra

No on-site servers. Cloud-native, auto-scaling, always available. Backed by Google infrastructure.

The return on investment is measured in **reduced unplanned downtime**, **lower mean-time-to-repair**, and the progressive shift from emergency maintenance to planned, predictive maintenance — all translating directly to production output and cost efficiency.